



TERM 2

NETWORKS

CHAPTER 4

CHAPTER OVERVIEW

Unit 4.1 Networks

Unit 4.2 Internet services

Unit 4.3 Internet connections

By the end of this chapter, you will be able to:

- Discuss a wide-area network.
- Explain what the internet is.
- Identify and explain various internet services.
- Identify and explain government internet services.
- Discuss the concepts of broadband and bandwidth.
- Explain throttling and shaping.

INTRODUCTION

The internet is one of the largest, most powerful and entertaining inventions ever made by humans. It has brought us a wealth of information, it has enabled us to conduct business from our homes, it connects millions of people and it provides an endless source of entertainment. But how exactly does it work? In order to answer this question, we will be taking a look at the basis of the internet – i.e. the network.

In this chapter, we will look at a wide-area network (WAN), as well as the best-known wide-area network, the internet. We will discuss how to obtain an internet connection and give some examples of internet services. Finally, we will explain broadband and bandwidth, and how they are affected by throttling and shaping.

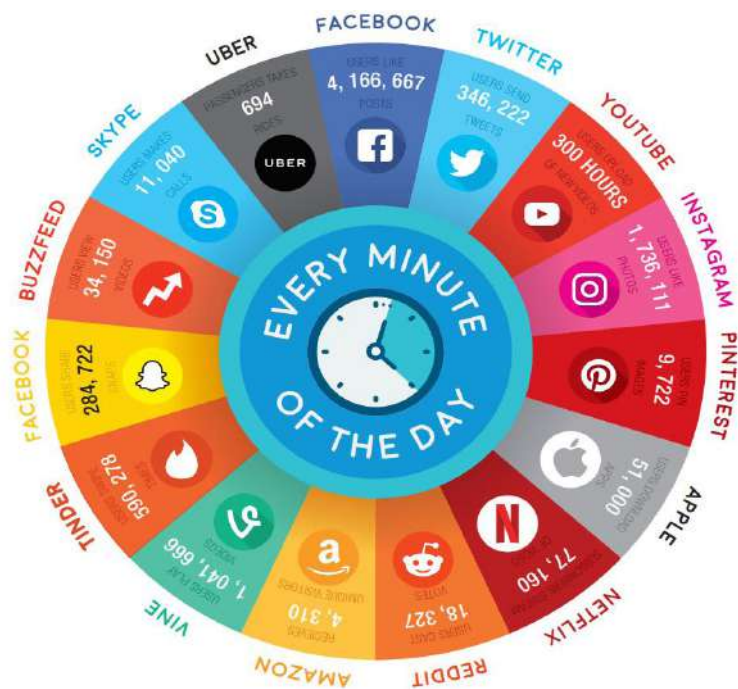


Figure 4.1: A breakdown of internet usage every day

In previous years, you learned that a network is a series of independent computers that have been connected with either a physical, or a wireless connection. In Grade 10, you learned about home and personal area networks, and in Grade 11, you learned that a **local area network** (LAN) is a computer network that covers a small area, such as a house or office, and that the computers in the network share resources, such as internet connections, printers and server connections. The network is usually limited to a certain number of computers (between two and 25), but there is no absolute limit.

Networks are incredibly powerful tools that are used on a daily basis by businesses and private users. These uses include the following:

- Sharing data and information
- Sharing hardware
- Storing information
- Providing access to entertainment
- Providing access to services
- Connecting people

WIDE-AREA NETWORKS

A **wide-area network** (WAN) is a network that covers a large geographical area, such as a neighbourhood, a city, a country, or even the world.

A WAN connects two or more LANs. A typical use of WANs in business is where different branches of a business are connected to share resources. This can be done through the use of satellites or telecommunication networks. Unlike LANs, many WANs are not private; instead, they are organised by communities or businesses. The following table shows some of the advantages and disadvantages of a WAN.

EQUIPMENT NEEDED TO CONNECT TO A NETWORK

To create a wired network, you need the following three types of networking equipment:

1. **A network adapter:** This is a piece of hardware that can be added to a computer and makes it possible to connect to a network. These days, most computers and laptops have a network adapter built into their motherboards.
2. **Network cables:** These cables are used to connect a computer to a LAN, or to connect one network to another network. If you are connecting directly to the internet, you may require a specific cable for the type of internet connection you are using, for example a DSL-enabled phone cable, or a fibre-optic cable.
3. **Hubs, switches or routers:** These are devices that allow multiple computers to connect to the same network. They can also be used to connect two networks to each other.

Something to know

Setting up a wired LAN versus a wireless LAN



<https://www.geeksonsite.com/faqs/how-to-setup-a-router/>

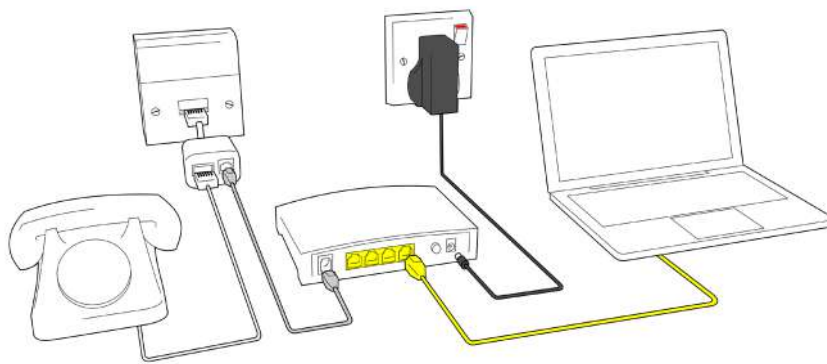


Figure 4.2: *Setting up a wired connection*

To create a wireless network, you only need two types of networking equipment:

1. Wireless network adapter
2. Wireless router or access point

Much like a network adapter, a wireless network adapter is a device that you can purchase and connect to your computer to access wireless networks. Most notebooks and smartphones have built-in wireless network adapters, while most desktop computers do not.

As with a wired network, you can use a router with wireless capabilities to connect all your devices in the same network. The router can connect all your devices to the internet. If you do not need internet access, you could also use a **wireless access point** (WAP). A WAP is like a hub for wireless devices. It allows wireless devices to connect to each other in a local network, but does not connect the local network to the internet.

THE INTERNET AS AN EXAMPLE OF A WAN

The internet is a computer network that consists of billions of connected devices, allowing people to share information, obtain entertainment, or chat to one another.

HOW DOES THE INTERNET WORK?

The internet is a global network of networks. It is made up of many different networks, in different countries around the world that are all connected together into one huge WAN.



Something to know

In 2018, there were more than 23 billion devices connected to the internet.

That is almost three internet-connected devices per person! By the year 2020, this number is expected to increase to 50 billion devices.



The following diagram shows how a user connects to the internet. Remember that the user can be a group of users in a network that connect to the ISP. There are also many different ISPs; each with their own network connected to the internet.



Figure 4.3: *How the internet works*

When the ISP gives its user access to the ISP's network, the users are connected to the internet. Users connect to their ISP via telephone lines, wirelessly, or through other standard networking methods. The information is transferred between computers using protocols. A **protocol** is a set of rules for the transmission of data between devices. Let's take a look at some of the following common protocols:

- **File transfer protocol (FTP):** Used to transfer and manipulate files on the internet
- **Hyper-text transfer protocol (HTTP):** An Internet-based protocol for sending and receiving web pages
- **HTTP secure (HTTPS):** Secure HTTP protocol; used mainly for e-commerce
- **Internet message access protocol (IMAP):** Used for email messages on the internet
- **Post office protocol version 3 (POP3):** Used by email clients to retrieve messages from remote servers
- **Simple mail transfer protocol (SMTP):** Used for email messages on the internet
- **Email client protocol:** is a standard method of exchanging information between email clients like Thunderbird or Apple Mail and email provider's servers like Gmail, Outlook, Yahoo, and vice versa
- **Voice over internet protocol (VoIP):** Used to transmit voice data (phone calls) over the internet.

You will learn more about some of these protocols in the following unit where we will get to discuss internet services.



Activity 4.1

1. Write down the correct answer for each of the following questions.
 - a. Which one of the following is NOT part of networking equipment?
 - A. Network cables
 - B. Network signaller
 - C. Hub
 - D. Router
 - b. An advantage of a WAN is that everyone on the network can access the same data and _____.
 - A. Videos and photos
 - B. Information
 - C. News
 - D. All of the above
 - c. A disadvantage of a WAN is that the setting up of it can be _____.
 - A. Difficult
 - B. Time consuming
 - C. Expensive
 - D. All of the above
2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 Protocol used to transfer and manipulate files on the internet	A. VoIP
2.2 Protocol for emails over the internet	B. IRC
2.3 Protocol to transmit voice data over the internet	C. HTTP
2.4 Protocol to receive emails from remote servers	D. IMAP
2.5 Protocol used to chat over the internet and also used for other forms of communication	E. REC
	F. POP3
	G. HTTPS
	H. FTP

3. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if they are false.
 - a. An extender is a piece of hardware that can be added to a computer and makes it possible to connect to a network.
 - b. The internet is the best-known LAN.
 - c. A WAN is a computer network that covers a small area.
 - d. To create a wireless network, you only need two types of networking equipment: a WAP and a wireless network adapter.
 - e. In modern computers, the network adapters are built into the motherboard.

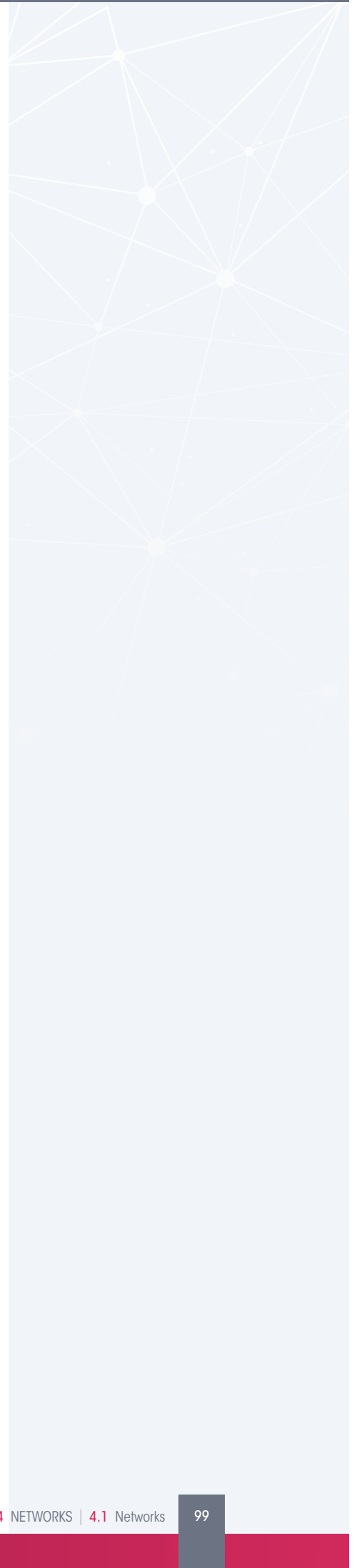




Activity 4.1

... continued

4. Answer the following questions.
 - a. What is a WAP and why is it important in a wireless network?
 - b. In your own words, describe what the internet is and why it is important in modern society.
 - c. What role does an ISP play when connecting to the internet?
 - d. Large networks have changed the way in which we share and store information. Name three additional advantages of large networks.
 - e. If you are looking to create a home network, what equipment will be needed? Will this change when you want to connect the network to the internet?
 - f. What are the advantages and disadvantages of using a wireless network instead of a wired network when connecting to the internet?
 - g. Give an example of where HTTPS protocol is used and how it differs from HTTP.



In the previous unit, you learned that you can access the internet via an ISP. Once you are connected to the internet, there is a range of services that you can use.

INTERNET SERVICES

Each type of internet service available is responsible for transferring a specific type of information. These services include the following:

- Communication services
- Information-retrieval services
- Web services
- World-wide web
- Video conferencing

In the following sections, we will be taking a closer look at some of these internet services, as well as their advantages and disadvantages.

INSTANT MESSAGING

Instant messaging, sometimes referred to as IM, is a term used to describe online chat programs that allow the user to send real-time messages over the internet. This includes text messages, as well as multimedia messages, such as pictures, voice notes and videos. Short, instant messages usually occur between two parties. However, you do have the option to have a group discussion where multiple users can participate.

ADVANTAGES AND DISADVANTAGES OF INSTANT MESSAGING

Instant messaging has played a big role in shaping the way in which we communicate on a daily basis. This is because of the wide variety of uses that this method of communication offers. However, it is not without fault. Here are some of the advantages and disadvantages of instant messaging.

ADVANTAGES OF INSTANT MESSAGING

- Allows you to chat in “real time” to other people who also have an IM client
- Messages are delivered immediately after being sent, if the person is online or is connected to the internet
- One can see when your message has been read
- It connects people regardless of where they are located
- People can speak to multiple people in virtual conference/groups, share ideas and reach conclusions.

DISADVANTAGES OF INSTANT MESSAGING

- Because it is immediate, you have no time to reflect on the message you are sending, unlike an email where you can review the draft before sending it
- Messages are not always saved
- Instant messaging takes away the face to face, personal experience that people have when they are speaking to someone in person.

EXAMPLES OF INSTANT MESSAGING SERVICES

There are a few main instant messaging services that you most probably use on a daily basis. Let's take a look at some examples of the most popular instant messaging services:

- **WhatsApp:** This is a popular instant messaging service that is used on smartphones and computers
- **Facebook Messenger:** This service is built into Facebook
- **iMessage:** This is Apple's built-in instant messaging service



Figure 4.4: WhatsApp is a popular instant messaging service

VOICE OVER INTERNET PROTOCOL (VOIP)

If you have ever been fortunate enough to travel overseas, you have most likely come to the realisation that phoning home is expensive. To alleviate this, internet companies have invented technologies that allow the user to make voice calls over the internet.

ADVANTAGES AND DISADVANTAGES OF VOIP

VoIP converts sound into data. The data is then transmitted over the internet to the recipient, where it is converted back to sound. Today, most online call services allow you to make video calls, enabling you to see the person (or people) to whom you are speaking. Since a large part of communication is non-verbal, being able to see a person's body language and facial expressions, while speaking to them, can be very useful. This is especially useful in a video conferencing situation, where there are multiple people connected to the same video call. In such a situation, being able to see which person is speaking allows you to communicate significantly more clearly. On the next page are some of the other advantages and disadvantages of VoIP.



ADVANTAGES OF VOIP

- Saves money on travelling to another country to attend a meeting.
- Audio and video information can be shared.
- Enables collaboration and can strengthen relationships between colleagues who work far away from each other.
- VoIP is much cheaper than using traditional telephonic calls.

ADVANTAGES OF VOIP

- It is expensive to set up and maintain.
- You need to have an active internet connection.
- Power failures will cause a break in communications.

EXAMPLES OF VOIP SERVICES

By now, you are most probably aware of the most-famous VoIP service, Skype. Skype has been around for more than a decade and since it is owned by Microsoft, it is integrated with Windows 10. Let's take a look at the following examples of VoIP services:

- **Google Hangouts:** This is a standard VoIP service provided by Google, which has recently been built into Android cellular devices.
- **WhatsApp:** This instant messaging application allows the user to make both standard and video calls over the internet.
- **Discord:** This application has become very popular for gamers, allowing them to talk to each other over the internet.
- **TeamSpeak:** Like Discord, TeamSpeak is used mostly by gamers to talk to each other over the internet.
- **Skype** is an example of a voice over Internet Protocol (VoIP) software application used for voice, video and instant messaging communications.



Figure 4.5: A Skype conference call

FILE TRANSFER PROTOCOL (FTP)

FTP, is a standard network protocol that is used to transfer files between a computer and the network server (usually over the internet). It does this by setting the rules for the way in which files can be uploaded and downloaded. For example, it can require you to use a username and password to access files on the server.





ADVANTAGES AND DISADVANTAGES OF FTP

No matter which business you are running, if you need to share large amounts of data internally, an FTP service might be the answer. Here are some of the advantages and disadvantages of using FTP.

ADVANTAGES OF USING FTP

- Easy to use (with the proper training)
- Multiple file directories can be sent at the same time
- File transfer progress is not lost if your connection fails
- File transfers can be scheduled at times that are best suited to your needs

DISADVANTAGES OF USING FTP

- Data, username and password are shared in plain text, which makes it easy for hackers to access information
- Requires training to use FTP effectively
- It is difficult to monitor mishandled data and track the source of the problem
- Not all FTP services encrypt your data, which makes it vulnerable to attack

EXAMPLES OF FTP SERVICES

While FTP was used quite commonly in the past, it is now used mainly in technical situations (for example, when uploading new files to a web server). Let's take a look at some of the examples of the most commonly used FTP servers:

- **FileZilla Server:** This is an FTP server available to Windows users.

GRID COMPUTING AND CLOUD COMPUTING

As technology advances, so do our requirements. It is no longer sufficient, nor economical to use your own personal computer to analyse and store complex data. An alternative was clearly needed. This answer was obtained on the form of cloud-based and grid computing.

- **Grid computing** is a computer model that has been connected to a network. It has the ability to process large volumes of data with the help of a group of networked computers that coordinate to solve a problem together.
- Cloud computing is the use of hardware or software on a server on the internet. It allows users to use software or save data without worrying about using up storage space.

ADVANTAGES AND DISADVANTAGES OF GRID AND CLOUD COMPUTING

Although cloud and grid computing offer huge benefits to users, they also have drawbacks. Here are some of the advantages and disadvantages of cloud and grid computing.

ADVANTAGES OF GRID COMPUTING

- It can help you solve larger, much more complex problems in a much shorter time span
- You do not have to buy more powerful hardware as you will make better use of existing hardware
- If one computer fails, the other computers will take over the work, which makes the system very reliable



GRID COMPUTING VS CLOUD COMPUTING

A Comparison You Wanted
to See



[https://www.youtube.com/
watch?v=PX89E8wQXQs](https://www.youtube.com/watch?v=PX89E8wQXQs)





Something to know

GIMPS (Great Internet Mersenne Prime Search):

GIMPS was formed in January 1996 to discover new world-record-size Mersenne primes. GIMPS harnesses the power of thousands of small computers like yours to search for these “needles in a haystack”.

DISADVANTAGES OF GRID COMPUTING

- Training is needed to make full use of grid computing
- May require multiple people in order for the grid system to function

ADVANTAGES OF CLOUD COMPUTING

- Since the cloud platform is managed and updated, it is much more reliable
- It can be used from any device capable of connecting to the internet

DISADVANTAGES OF CLOUD COMPUTING

- You may experience downtime if the cloud server is undergoing maintenance
- Even though cloud computing uses security measures to protect your data, there is always a risk of files being compromised
- You do not have full control over the platform as it is owned and managed by the service provider

EXAMPLES OF CLOUD AND GRID COMPUTING

Both cloud and grid computing help the user to distribute computing over a large area. Table 4.1 shows examples of the most commonly used cloud and grid computing software.

In the ideal grid computing system, every resource is shared, turning a computer network into a powerful supercomputer. With the right user interface, accessing a grid computing system would look no different than accessing a local machine's resources. Every authorised computer would have access to enormous processing power and storage capacity.

- **World Community Grid:** World Community Grid's mission is to create the largest public computing grid benefiting humanity, which is funded and operated by IBM. Using the idle time of computers around the world, World Community Grid's research projects have analysed aspects of the human genome, HIV, dengue, muscular dystrophy, and cancer.
- Scientific research, helping scientists around the world to analyse and store massive amounts of data.
- **SKA (Square Kilometre Array):** The SKA is a worldwide network of radio telescopes that will consist of over one million square metres of collecting area. This will generate an enormous amount of data will be processed by an enormous computing grid consisting of more than half a million computing cores (processors) distributed around the globe and linked by a powerful network.

Table 4.1: Commonly used cloud and grid computing software

CLOUD COMPUTING	GRID COMPUTING
1. Amazon Web Services (AWS)	1. Berkeley Open Infrastructure for Network Computing
2. Microsoft Azure	2. Advanced Resource Connector
3. Google Cloud Platform	3. g-EclipseTechnology Project





Figure 4.6: Examples of cloud and grid computing software

CLOUD-BASED FILE SHARING

Since the advent of the internet, the technology industry has been steadily moving away from local storage to remote, server-based storage and processing—what is known as the cloud. We’ve rounded up the best cloud storage and file-sharing and file-syncing services to help you decide which are right for you.

ADVANTAGES AND DISADVANTAGES OF FILE SHARING

Cloud-based file sharing allows the user to easily store and share files, documents, photos and videos on the internet. However, it also has its drawbacks. Here are some of the advantages and disadvantages of using cloud-based file sharing.

ADVANTAGES AND DISADVANTAGES OF CLOUD-BASED FILE SHARING

- It is very easy to use as you can just drag and drop files between your computer and the file-sharing service
- It can save data and bandwidth, as you can share the file with multiple people instead of having to mail it to each of them
- You can access the files with any device that is capable of connecting to the internet
- It is very convenient as it reduces the time and effort it takes to share information with multiple people

DISADVANTAGES OF CLOUD-BASED FILE SHARING

- The files might not be very secure as multiple people will have access to them
- People might misplace or delete some files by mistake
- You need an internet connection to access the files
- Some services might require software to use, which will require you to download the software before being able to access the files

EXAMPLES OF CLOUD-BASED FILE SHARING

To use cloud-based file sharing, you will need to use a service provider. The service provider is responsible for providing you with a service that is highly available and provides backups and file recovery in a timely manner. To help make this decision easier, let’s look at some of the most commonly used cloud-based files-sharing services.





Table 4.2: *Most commonly used cloud-based file-sharing services*

FILE-SHARING SERVICE	DESCRIPTION	IMAGE
Dropbox	This is the most widely used file-sharing service, as of May 2018	
MediaFire	This is a file-sharing service that can be used with Windows, MacOS and Linux	
Google Drive	This is a file-sharing service provided by Google	
iCloud	This is a file-sharing service provided by Apple	
Microsoft OneDrive	This is a file-sharing service provided by Windows and MacOS	

These services provide seamless access to all your important data—Word docs, PDFs, spreadsheets, photos, and any other digital assets—from wherever you are. You no longer need to be sitting at your work PC to see your work files. With cloud syncing you can get to them from your smartphone on the train, from your tablet on your couch, and from the laptop in your hotel room or kitchen. If you don't yet have a service for storing and syncing your data in the cloud, you should seriously consider one. Which you choose depends on the kinds of files you store, how much security you need, whether you plan to collaborate with other people, and which devices you use to edit and access your files. It may also depend on your comfort level with computers in general. Some services are extremely user-friendly, while others offer advanced customisation for more experienced technophiles.

STREAMING

Broadcast, cable and satellite television have been responsible for supplying news and entertainment to households for many years; however, this may be coming to end. With the advances made in the internet, a more diverse service has become available – i.e. streaming.

Streaming is a service that allows the user to view live events, series, movies and sporting events over the internet. This includes streaming to your smartphone, smart television or computer.



UNDERSTANDING VIDEO STREAMING



<https://www.youtube.com/watch?v=AeJzoqtuf-o>





ADVANTAGES AND DISADVANTAGES OF STREAMING

Streaming has made it very easy to catch up on series, view sports and keep up to date with the news. But like all other entertainment services, it is not without a downside. Here are some of the advantages and disadvantages of streaming.

ADVANTAGES OF STREAMING

- You are in control of what you watch and when you want to watch it
- Streams can be paused, rewind and resumed as you please
- It does not require you to have the shows on your computer and, therefore, saves storage space

DISADVANTAGES OF STREAMING

- You need an active (fast) internet connection in order to stream
- Streaming may use a lot of data
- The luxury of streaming may result in health problems as it is easy to spend many uninterrupted hours watching series and shows

EXAMPLES OF STREAMING SERVICES

Over the past few years, streaming has become more popular and widespread. Let's take a look at some of the examples of streaming services that you might find helpful:

- **Twitch:** This is a streaming platform dedicated to gamers. Gaming tournaments and "Let's Play" are some of the most popular streams on this platform.
- **Netflix:** This is a platform dedicated to series and movies. Netflix has become very popular as it releases a whole season of a series at once, allowing the user to dedicate a block of time to binge watch the series.
- **Showmax:** This platform was released in South Africa just before Netflix. Showmax allows you to stream some titles that are not available on Netflix, including Afrikaans movies and series.



Figure 4.7: *Twitch, the streaming platform for gamers*

GOVERNMENT INTERNET SERVICES

Government institutions are known to have queues in which you could wait for hours. Thanks to the internet, many government institutions now have quick and easy solutions. Government internet services are internet services that allow users to skip queues by using government websites, from the comfort of their homes, to conduct business, pay bills and obtain information.





ADVANTAGES OF GOVERNMENT INTERNET SERVICES

More and more people are beginning to use government internet services as it saves them a lot of time and effort. This, however, is not the only advantage. Let's take a look at some other examples of how government internet services can make your life easier.

PAYING YOUR TELEVISION LICENCE

You can pay your TV Licence online by using your credit card details, or online at your own banking website/banking app using the TV Licenses account details, or go to the EasyPay website for a virtual payment solution.

ELECTORAL INFORMATION

One of the advantages of living in a democratic country is that you are able to vote for the people you want to be in charge of your city and country. To do this, you need to know:

- Who the people and parties running for office are
- If you are registered to vote
- Where you should go to cast your vote
- What the results of the election were

This information and more can be obtained from the various government websites, which you can access through the internet. One such website is the "Elections" website (www.elections.org.za), which you can use to confirm if you are registered to vote.

SUBMITTING TAX RETURNS

Tax season is a stressful time of the year for every person involved, especially when you get to the SARS office and the queues are so long. Sometimes you need to return another day just to finish filing your taxes. Fortunately, SARS has come up with the solution of e-filing.

Electronic filing (commonly known as e-filing) gives users a free, simple and secure way to do tax returns from the comfort of their homes. It is available 24 hours a day and allows users to see their tax status, change their information and make any payments required. You can submit tax returns via e-filing on the SARS website (www.sarsefiling.gov.za).

SMART ID AND PASSPORT APPLICATIONS

Applications for Smart IDs and passports can be done online. This is done via the eHomeAffairs website. Instead of standing in long queues at Home Affairs Offices users can complete an online application, upload supporting documents, make payments and schedule a time to collect the documents from a Home Affairs enabled bank.

ONLINE TERTIARY APPLICATIONS

The Department of Higher Education and Training runs an online Central Applications Clearing House (CACH). This is open in January and February each year. Students who have not succeeded in finding a place at a tertiary institution can submit an online application via the CACH and they will be assisted to find a place at a tertiary institution.





Activity 4.2

1. Write down the correct answer for each of the following questions.
 - a. Which one of the following is NOT an example of FTP services?
 - A. Pure FTPd
 - B. FileZilla Server
 - C. ProFTPd
 - D. FT protocol
 - b. Which one of the following is an example of instant messaging?
 - A. Wikipedia
 - B. Facebook
 - C. Telegram
 - D. YouTube
 - c. When you are in control of what you watch and when you want to watch it.
 - A. Cable
 - B. Broadcast
 - C. Streaming
 - D. Satellite television
2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 Example of cloud-based file sharing	A. VoIP
2.2 Standard network protocol that is used to enable your computer to transfer files between the computer and the network server	B. SARS
2.3 May use a lot of data when watching a video	C. HTTP
2.4 An example of cloud computing	D. Microsoft Azure
2.5 Where you pay your taxes	E. REC
	F. Dropbox
	G. Streaming
	H. FTP

3. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if they are false.
 - a. Netflix is more popular than M-Net.
 - b. Showmax is an example of a streaming service.
 - c. Electronic payments give users a free, simple and secure way to do their tax returns from home.
 - d. Instant messaging allows the user to make voice calls over the internet.
4. Answer the following questions.
 - a. What is the difference between cloud computing and grid computing?
 - b. Explain e-filing and give one advantage thereof.
 - c. Why would people want to use government internet services? Mention a government institution that uses the internet.
 - d. List four advantages, one disadvantage and three examples of instant messaging.
 - e. Give two examples of apps on a smartphone that use the following internet services:
 - i. VoIP
 - ii. FTP
 - iii. Cloud computing and grid computing
 - iv. Cloud-based file sharing
 - v. Streaming



4.3 Internet connections

Throughout this chapter, we have looked at some of the wonderful services that you can access through the internet. However, in order to access these services, you need an internet connection. In this section, we will be taking a look at how you can connect to the internet and which internet connection is best for your purpose. In South Africa, you need to pay an ISP, such as MWeb or WebAfrica, or a **mobile network provider** (MNP), such as Vodacom or MTN, a monthly fee in order to access the internet.

BROADBAND

In the early days of the internet, you had to use a modem that allowed you to connect to the internet through dial up. This type of connection had many drawbacks; it was slow, it required you to dial up each time you wanted to connect to the internet, and it would disconnect you when you used your phone. This all changed with the invention of broadband.

Broadband is a high data-rate connection that allows the user to send data over multiple channels. This makes the connection faster and more reliable than the old dial-up method. Examples of broadband include ADSL, 3G and fibre. It is important to note that not all broadband connections are available in all areas. When you decide on getting either a wired or a wireless connection, you should consider the coverage and availability of the various connections in your area. You can get this information from your service provider, or you can look at a network coverage map on a service provider's website.



Figure 4.8: The green indicates areas with internet coverage

WHAT TYPE OF INTERNET CONNECTION DO YOU NEED?

When choosing an internet connection, you need to consider the following four factors:

1. Speed of the internet
2. Data cap of your connection
3. Cost of the internet
4. Equipment you will need to connect to the internet



INTERNET SPEED OR BANDWIDTH

When choosing an internet connection one needs to look at the following:

- Speed of the internet connection
- Type of connection one can afford

Bandwidth refers to the amount of data that can be transferred over a network in a specific time. Higher the bandwidth the more data can be transferred in less time. Downloading refers to the transfer of data from the internet to your computer. This includes downloading files, for example, videos, as well as browsing the internet. Uploading: refers to the transfer of data from a computer to the internet, for example, uploading a video to YouTube.

DATA CAPS AND BUNDLES

A **data cap** (bandwidth cap) is a service provider-imposed limit on the amount of data transferred by a user account at a specified level of throughput over a given time period, for a specified fee. The term applies to both home Internet service and mobile data plans.

Data caps are usually imposed as a maximum allowed amount of data in a month for an agreed-upon charge. As a rule, when the user exceeds that limit, they are charged at a higher rate for further data use. However, the provider may not charge overage but instead throttle the users' transfer rate per second beyond the limit. For example, a mobile user paying for a 4G plan may be downgraded to 3G for mobile data beyond their data cap.

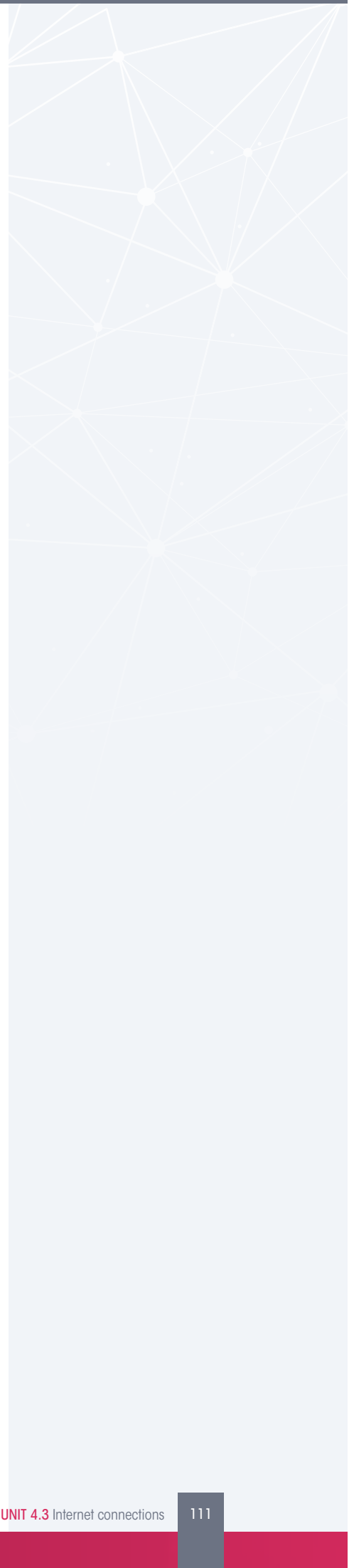
An ISP might impose a data cap when a customer shares access with many users, employs file-sharing software or otherwise breaks terms of use. Service providers also sometimes impose data caps when they have a monopoly in a given area.

Data caps result in lower rates of access to online materials, especially multimedia and streaming content which is a problem for content creators and their advertisers. Data caps are also uniformly unpopular with customers. As a result, service providers often refer to data caps by other names such as fair usage or fair access policies, usage based billing or vaguely as band caps.

A data bundle gives access to the internet. To surf we use internet data which is measured in bytes – with data bundles it doesn't matter how many minutes you stay connected, but rather how many bytes you use. It's a cheaper yet a convenient way to connect to the internet. Whether you just want to surf or make downloads it's still an ideal choice. It works more or less the same way we surf using airtime on our cell phones.

FAIR-USAGE POLICY

A **fair-usage policy** (FUP) (also called a reasonable-usage policy, or an acceptable-usage policy) is a data limit placed on "uncapped" internet accounts. According to this policy, your internet account will not be capped or limited unless you use too much data, in which case your internet access may be slowed (throttled) to the point where it is unusable. Many uncapped internet accounts have strict fair-usage policies that will catch unsuspecting users off guard.





INTERNET COSTS

The third thing to consider is the cost of the internet. Unfortunately, in South Africa, internet is still an expensive luxury item for most people. In general, a fast internet connection with a large data cap will be more expensive than a slow internet connection with a low data cap.

EQUIPMENT NEEDED

The final thing to consider is what equipment you will need in order to connect to the internet. This includes four basic components:

1. Computer
2. Internet connection channel
3. Internet communication device
4. An ISP

WHAT IS THROTTLING?

Bandwidth throttling is a purposeful slowing of available bandwidth. In other words, and in general, it's an *intentional* lowering of the "speed" that's typically available over an internet connection. Bandwidth throttling can happen at various places between your device (like your computer or smartphone) and the website or service that you're using over the internet.

WHY WOULD ANYONE WANT TO THROTTLE BANDWIDTH?

You as the user of an internet connection or service rarely benefit from bandwidth throttling. Very simply, bandwidth throttling means limiting how fast you can access something when online. For example, an ISP might throttle bandwidth during certain times of the day to decrease congestion over their network, which lowers the amount of data they have to process at once, saving them the need to buy more and faster equipment to handle internet traffic at that level.

Another reason a service provider might throttle bandwidth is to provide a way for users to avoid the throttling by paying for a more expensive service that doesn't limit bandwidth. In other words, the bandwidth throttling might just be an incentive to encourage heavy users to upgrade their plan.

WHAT IS SHAPING?

Traffic shaping is used for a number of purposes:

- Time-sensitive data may be given priority over traffic that can be delayed briefly with little-to-no ill effect.
- A large ISP (Internet service provider) may shape the traffic of an independent reseller.
- In a corporate environment, business-related traffic may be given priority over other traffic.
- An ISP may limit bandwidth consumption for certain applications to reduce costs and create the capacity to take on additional subscribers. This practice can effectively limit a subscriber's "unlimited connection" and is often imposed without notification.
- Traffic shaping could be an integral component of the proposed two-tiered Internet, in which certain customers or services would get traffic priority for a premium charge.





Activity 4.3

1. Write down the correct answer for each of the following questions.
 - a. Which one of the following is NOT an ISP?
 - A. MWEB
 - B. M-Net
 - C. Afrihost
 - D. Webafrica
 - b. Which one of the following is NOT an example of broadband?
 - A. 3G
 - B. ADSL
 - C. 4K
 - D. Fibre
 - c. Which one of the following is NOT an option for fast internet users whose speed is more than 20 Mb/s?
 - A. 56 k
 - B. VDSL
 - C. LTE
 - D. Fibre
2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 An internet connection without limited data	A. ADSL
2.2 Connection option suited for 4 Mb/s download speed	B. MTN
2.3 ISP with fibre option	C. 56K
2.4 What you pay to get internet access	D. LTE
2.5 Mobile network provider	E. ISP
	F. Capped
	G. Uncapped
	H. Afrihost

3. Say if the following statements are TRUE or FALSE. Correct the underlined word(s) if they are false.
 - a. Everything you do on the internet affects data.
 - b. Axxess is not an ISP for fibre connections.
 - c. You need an ONT device if you want to use DSL internet.
 - d. Speed of internet is not important when choosing an internet connection.
 - e. High-quality videos use about 1 Tb/h of data.
4. Answer the following questions:
 - a. What is the difference between uploading and downloading?
 - b. What is broadband?
 - c. What is the difference between a shaped and an unshaped internet connection?
 - d. What four things must you consider when choosing an internet connection?
 - e. Explain the FUP.





REVISION ACTIVITY

... continued

QUESTION 1: MULTIPLE CHOICE

- 1.1** What type of network is the internet? (1)
A. LAN
B. PAN
C. WAN
D. WHAN
- 1.2** Which of the following protocols sends web pages? (1)
A. IMAP
B. HTTP
C. VoIP
D. IMAP
- 1.3** Which of the following is NOT a service provided by ISPs? (1)
A. Instant messaging
B. World-wide web
C. Skype
D. Broadband
- 1.4** Which of the following is an example of an FTP? (1)
A. FileZilla Server
B. Amazon Web Services
C. MediaFire
D. Twitch
- 1.5** Which of the following is an advantage of government internet services? (1)
A. You can pay your water bills online
B. You can communicate through video conferencing
C. You can stand in shorter queues at SARS
D. You can find out where to pay your TV licence
- [5]

QUESTION 2: TRUE OR FALSE

Indicate if the following statements are TRUE or FALSE. Correct the statement if it is false. Change the underlined word(s) to make the statement true.

- 2.1** HANs can can be public and private. (1)
- 2.2** A WAP can be used to create a wireless network. (1)
- 2.3** FTP is used to connect computers in a network. (1)
- 2.4** Google Drive is Google's instant-messaging platform. (1)
- 2.5** 4G connections are a new technology of broadband. (1)
- [5]

... continued





REVISION ACTIVITY

... continued

QUESTION 3: MATCHING ITEMS

Choose a term or concept from Column B that matches a description in Column A.

(5)

COLUMN A	COLUMN B
3.1 Maven can use SARS e-filing to submit his tax returns	A. Uncapped data B. FTP C. VoIP D. ISP E. Government internet service F. Capped data G. Cloud computing
3.2 Carleen uses Skype to talk to her family while she is teaching English in Madagascar	
3.3 A method of sending documents and videos between computers over a network server	
3.4 Telkom provides Zoey with the tools she needs to send WhatsApp messages to her friends with her smartphone	
3.5 To save money, Darrel limits the amount of data his family can use each month	

[5]

QUESTION 4: CATEGORISATION QUESTIONS

Look through the following advantages and disadvantages, and indicate to which internet service each belongs.

(5)

ADVANTAGE OR DISADVANTAGE	INTERNET SERVICE
4.1 Since the cloud platform is managed and updated, it is much more reliable	
4.2 You are in control of what you watch and when you want to watch it	
4.3 Audio quality depends on the quality of your internet connection	
4.4 Maintaining multiple computers may become expensive	
4.5 The files might not be very secure as multiple people will have access to them	

[5]

... continued





REVISION ACTIVITY

QUESTION 5: SCENARIO-BASED QUESTIONS

Your cousin is moving out of the house soon and has started investigating connectivity options. She is considering the following packages:

PACKAGE 1	PACKAGE 2	PACKAGE 3
Uncapped ADSL Free Data after hours (24:00 – 08:00) Up to 2Mbps Semi-shaped	15GB 3G / LTE on Cell C Data only SIM No out-of-bundle rates 3 month Data Rollover	200GB Cap Fibre Up to 4Mbps Unshaped data No contracts

- 5.1** What does the term *uncapped* for package 1 mean? (1)
5.2 Explain why you would never be throttled on package 2 or 3? (1)
5.3 What does it mean when your internet connection is shaped? (2)
5.4 Which one of these packages has the highest speed? (1)
5.5 Refer to package 2: What does out-of-bundle rates refer to? (1)

Nkosi has to hold Skype conferences for his company for the next month. Recently, the company has been having serious problems with big clients; therefore, an emergency meeting can occur at any time, which means he must make sure he is readily available.

- 5.6** What should Nkosi consider when buying an internet connection for the next month? (4)
5.7 What internet specifications do you think would be best for him in this situation? Give a suggestion for each factor. (4)
5.8 Compare the difference between capped data and uncapped data. (4)
5.9 Explain to him what the purpose of a WAP is. (2)

[14]

TOTAL: [40]

AT THE END OF THE CHAPTER

NO.	CAN YOU ...	YES	NO
1.	Explain what a WAN is?		
2.	Explain what the internet is?		
3.	Identify and explain various internet services?		
4.	Identify and explain government internet services?		
5.	Discuss the concepts of broadband and bandwidth?		
6.	Explain throttling and shaping?		





SOCIAL IMPLICATIONS OF NETWORKS

CHAPTER 5

CHAPTER OVERVIEW

Unit 5.1	Computer-related crimes
Unit 5.2	Security issues
Unit 5.3	Security and safeguards
Unit 5.4	Impact of technology on the global community

By the end of this chapter, you will be able to:

- Discuss computer-related crimes, attacks and frauds.
- Describe security problems.
- Identify how to avoid security threats.
- List and describe safeguards against cyber attacks.

INTRODUCTION

The internet has made it easier than ever to obtain, share and store information on a platform that you can access from around the world. However, it has also created a new gateway for people with malicious intent to commit crimes against others.

In a world where your computer or smartphone can be used to transfer money with online banking, obtain ID documents and passports, it is important to ensure that you and your information are protected and secure. In this chapter, we will look at some examples of computer-related crimes, attacks and frauds. We will also discuss some security problems that you need to be aware of and how to avoid these threats. We will also give advice on how to protect yourself against cyber attacks.



Figure 5.1: *Cyber security is always important*

Computer-related crimes, or cybercrimes, have become more frequent as new methods are discovered to infiltrate your computer. From using something as simple as a virus, to more complex attacks such as phishing, cybercriminals aim to access people's computers and their passwords. These can then be used to access the users' bank accounts, steal their identities, or obtain their personal information. Cybercriminals can also use their computers to send spam emails and add unwanted advertisements to every website that the user visits. It is for this reason that computer users need to understand the basics of computer crimes, as well as how to use computers and the internet safely.

TYPES OF COMPUTER-RELATED CRIMES

Computer-related crimes range from physical crimes to crimes committed over the internet. In the sections that follow, we will look at the different types of crime.

THEFT OF COMPUTER HARDWARE

Theft of computer hardware refers to the theft of physical hardware components, which includes internal components, such as RAM and graphics cards, external components such as a keyboard and mouse, as well as the theft of your laptop or smartphone.

Computer hardware is very expensive and is, therefore, very important that you protect yourself against hardware theft. Let's take a look at some general guidelines on how to protect yourself:

- Never leave your laptop or mobile device unattended in a public place.
- Use passwords as a security method.
- Use a cable to lock your equipment to a table or desk in the office.

SOFTWARE THEFT

Software theft refers to the use and distribution of software obtained by illegal methods. This includes stealing a CD or DVD that contains software, as well as downloading illegal software from the internet (piracy).

Software theft and piracy are crimes in many countries, including South Africa. Therefore, it is important to make sure that you do not partake in these activities. Let's take a look at some guidelines to help you stay clear of software theft:

- Do not copy, duplicate or distribute any software without a copyright licence.
- Do not download and use illegal software from the internet.
- Do not lend your software so that an illegal copy can be made.

BANDWIDTH THEFT

Bandwidth theft refers to the use of bandwidth without paying for it. People steal bandwidth in many ways, which includes stealing usernames and passwords and connections that are not owned by them. Bandwidth theft can also occur when one is connected to an unprotected network.

Something to know

Piracy is one of the most widespread computer-related crimes in modern society. This is due to the widespread illegal downloading of music, TV series and movies.

Precautions include:

- Having good password policies.
- Being careful when installing free software from the internet, as it might be malware.

The second type of bandwidth theft is hotlinking. **Hotlinking** is when someone takes a file or video from his or her website and links it to your website. This is problematic as each time a user clicks on the link in order to download the file or view the video, your bandwidth will be used.

IDENTITY THEFT

Identity theft refers to stealing data that can be used to pretend that a person is someone else. Identity theft can range from stealing any personal information such as your name, your address, your favourite food or more official information such as your ID number, telephone number, etc. The stolen information can be used to take over your accounts on websites like, Twitter, Instagram, etc, so as to post messages or use these services whilst pretending to be you. Personal information, such as identity numbers, can be used to impersonate a person (i.e. steal his or her identity). This allows the thief to open accounts, take out loans and buy things in that person's name.

To avoid these problems, do not share your sensitive information unnecessarily and protect your information using a secure password. While sharing information is part of using the internet, be selective with the websites on which you choose to share your information. If you use your email address for every website you visit, do not be surprised if you start receiving a lot of spam emails!



Figure 5 2: *A hacker using someone else's details to hack into his or her account*



MISUSE OF PERSONAL INFORMATION

Information about you has value. For example, if you use Google to search for the rules of the computer game “Defence of the Ancients” (“Dota”) and then watch YouTube videos about how to play it, your web browser and search engine may record this information. Without you telling them, they have now learned that you like the game and might be interested in playing it. This information is very valuable to people who make games or gaming hardware. Google then takes this information (and all the information it records about the billions of people who use its software) and sells it to advertisers. The next time you open a website, you suddenly see an advertisement for a brand-new gaming mouse in your web browser!

THEFT OF TIME AND SERVICES

Theft of time refers to a person receiving payment from his or her employer for work that the employee did NOT do. This is usually because the person is busy with non-work-related activities during work time. Theft of services refers to the person using company equipment, such as a computer or the internet, for personal matters.

INTERNET-RELATED FRAUD

A **scam** is a fraudulent scheme used by a malicious person to obtain money or something of value from someone else.

Table 5.1: *Examples of internet scams*

TYPE	DESCRIPTION	PROTECTION
Phishing and email spoofing	Phishing and email spoofing attacks try to obtain sensitive information (such as usernames, passwords and banking details) by sending emails to users that look like official emails. These emails will either directly request the sensitive information, or redirect users to an official-looking website where their information will be stolen.	Never send sensitive information, such as usernames and passwords over email. No company will ever ask for this information via email or phone. Make sure the email address and domain name are correct for any email that looks suspicious.



Figure 5.3: *Internet phishing*





CROWD FUNDING

Crowd funding is a way for individuals and organisations to raise money from a large number of people. It can be an alternative to using a bank or investment firm. Below are some examples of crowd funding:

DONATION CROWD FUNDING

Donation crowd funding is used to raise money for a cause. It's not just charities who can do this – groups and individuals can also start crowd funding projects, for example to pay for a community centre or a child's surgery.

REWARD CROWD FUNDING

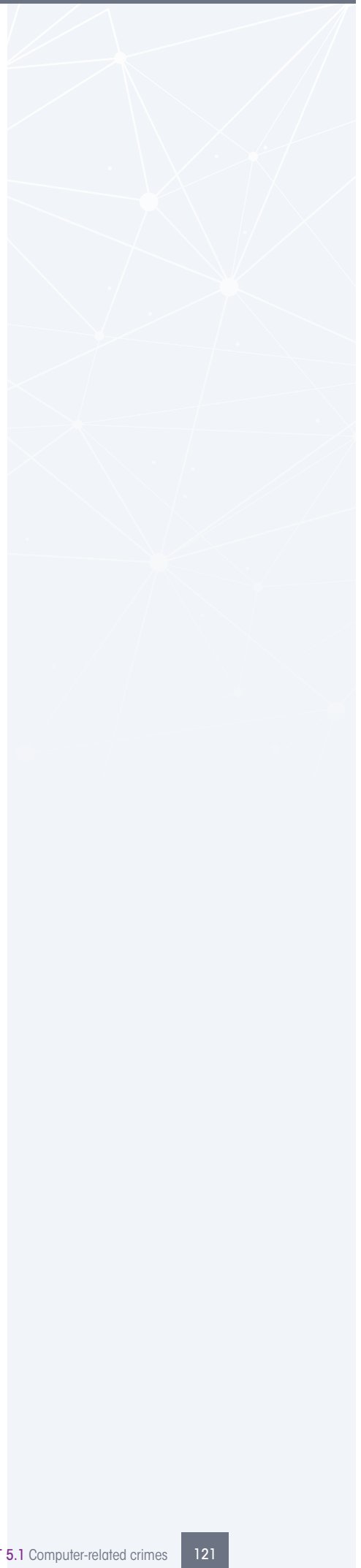
This involves funding a project for a set reward. For example, you might give money to help a band record its music and get their album in return. The reward may vary depending on how much money you give. For example, if you gave R20 you might get the album, but if you gave R50 you might get the album and a t-shirt.

LOAN BASED CROWD FUNDING

This is also known as “peer-to-peer” (P2P) and “peer-to-business” (P2B) lending. You'll lend a certain amount of money, which should be repaid to you with interest over time. So, the idea is you get back more than you lent.

INVESTMENT CROWD FUNDING

Investment crowd funding (also called “equity crowd funding”) typically involves buying part of a company. The part you own is called a “share”. You'll usually be one of many shareholders, along with other crowd funders, investment firms and people working at the company.





Activity 5.1

1. Write down the correct answer for each of the following questions.
 - a. When a thief knows and sells the information about what you like and do not like, or do and do not do, what do we call this type of identity theft?
 - A. Personal information
 - B. Financial information
 - C. Social media information
 - D. Interests
 - b. Which of the following is NOT an internet scam?
 - A. Phishing
 - B. Cryptocurrencies
 - C. "Nigerian letter"
 - D. "Congratulations, you have won"
 - c. You have received an email from your bank saying that the bank is converting to an electronic system and asks you to type in your banking details so that it can be saved onto their system. Which of the following scams fit this scenario?
 - A. Phishing
 - B. Cryptocurrencies
 - C. "Nigerian letter"
 - D. "Congratulations, you have won"
2. Choose a term or concept from Column B that matches the description in Column A. Write only the letter next to the question number.

COLUMN A	COLUMN B
2.1 World-wide illegal sharing of music, TV series and movies	A. "Congratulations, you have won"
2.2 This scam requires the target to make a financial investment with promises of high returns and little risk	B. Phishing
2.3 Internet-related fraud	C. Social media information
2.4 Type of computer-related crime	D. Care-share
	E. Piracy
	F. Denial-of-service
	G. Crypto

3. Answer the following questions:
 - a. What steps can you take to protect yourself from computer hardware theft?
 - b. Explain bandwidth theft.
 - c. What steps can you take to protect yourself from bandwidth theft where a third party gains access to your internet connection?
 - d. Explain how other people can misuse your personal information.
 - e. Describe two examples of internet scams.



5.2 Security issues

One of the disadvantages of networks is that other people can sometimes get unauthorised access to computers via the network. Security breaches can occur in LANs, but there is a bigger threat in WANs, such as the internet.

INTERNET ATTACKS

Internet attacks or cyber attacks refers to an attack on the infrastructure of the web itself. Examples include the following:

- **Denial-of-service (DOS/DDOS):** The use of compromised computers to flood a server with requests for information. The attack can be regarded as successful if the server becomes so overwhelmed that normal users cannot connect to the system.
- **Cyberterrorism:** Attacks that only occurs in parts of a country's IT infrastructure.
- **Spoofing attacks:** Spoofing attacks work by masking the attacker and making it seem as if he or she is a legitimate user. This is done by using false data to connect to the network.
- **Sniffer attacks:** Sniffer attacks works by checking the data packets sent over the network. Hackers use sniffer applications to analyse the network and access all unencrypted data being transmitted.

MALWARE

In our modern society, computers and smartphones are used to do everything; from transferring money with online banking, to storing personal information on the cloud. It is, therefore, very important that computer users understand the basics of computer malware, and know how to use computers and the internet safely. Table 5.2 gives some information on the most common threats and how best to avoid them.

Table 5.2: Common security threats

TYPE	DESCRIPTION	PROTECTION
Pharming	A scamung practice where a user is automatically redirected to a fake website even if they type in the correct URL of the official site.	• Take note of redirection warnings • Check whether the site is secure • Make sure that your anti-soyware software is up to date
Click-jacking	Click jacking is an attack in which a site appears to do one thing but your click are actually redirected to do something else	• Add ons can be installed on the browser • Sever programs can be installed to protect users on a network
Spoofing	The changing of an email header so that the email appears to come from a different source	• Make sure the email comes from a trusted sender
Spyware	Any technique or software that tries to monitor and track the way you use your computer, once it has done this, your information is sent to a third party.	• Install anti-spyware software. • Do not fall victim for any internet hoaxes



Something to know

Malware is a term used to describe software that can be used to cause harm to the user's computer or computer network, without the user's knowledge. This includes viruses, Trojans, worms, bots and zombies. Malware programs normally function by causing your computer to perform unwanted actions, such as sending emails and messages to help spread the infection, giving the hacker personal information stored on your computer, or allowing the hacker to take control of your system.



Something to know

A pop-up is a form of advertising normally found on the internet. The advertisement consists of a small window that suddenly appears (pops up) with a message asking the user to click, **in order** to claim a prize, gain information on a product, or go to a website for a special offer. Although some of the pop-ups are safe and will provide you with the promised information, there are many that are used in order to spread malware, such as spyware and adware.

Table 5.2: Common security threats

TYPE	DESCRIPTION	PROTECTION
Keylogger	A type of software that records your keystrokes as you type in an attempt to discover your personal information	• Make use of a firewall • Install a password manager. • Keep your system updated • Consider anti malware software
Computer virus	A program that is written to disrupt the normal functioning of a computer. Most viruses activate when you open or run an infected file	• Install anti-virus software and keep it updated • Scan email attachments and portable storage devices • Do not open email attachments or run programs from untrusted sources
Computer worm	Malware that is able to distribute itself over a network. It is able to reproduce itself. Usually distributed via email, without a person having to run an infected program.	• Install anti virus software • Do not open unrecognised or suspicious emails
Trojan	A disruptive program disguised as a useful application. It relies on one to activate it before it can operate	• Do not run any programs unless you are sure it is safe to do so
Ransomware	Malware that makes data unavailable until you pay the hacker a ransom fee	• Install anti-virus software and keep it updated.
Adware	Software that downloads itself onto your computer and bombards your computer with adverts that pop up whenever you are connected to the internet	• Install ant-malware software • Do not click on any unknown links
Scams	A message or article intended to deceive or defraud others	• Check sources for reliability • Do not forward a scam to others • Never take any actions suggested in a scam message

BOTS

A **bot** (short for robot) is a software application that allows the user to automate tasks and actions. On the internet, they are normally used to analyse and gather file information. An example of this is web crawlers. These bots gather, analyse and index data that can be used by search engines, such as Google.

While bots can be used to make the user's life easier, they often come in the form of malware. **Malware bots** function by infecting the user's computer and automatically performing malicious actions, such as gathering keystrokes, obtaining financial information, gathering passwords, or opening back doors on the infected computer.





ZOMBIES

Malware bots also have the capability to completely take over the user's computer, allowing the hacker to take control. When this happens it is known as a **zombie**. Hackers can use zombie computers for a wide variety of applications; including launching DoS attacks and sending spam emails **in order** to spread the malware.

RIGHT TO ACCESS VS RIGHT TO PRIVACY

We know by now that computer crimes depends on access to data. For one to prevent fraud and identity theft one needs to keep their data private and one needs to protect it.

The important part here is to know who has the right to access data, versus your right to keep your data private. Below are a few examples to look at when it comes to the right of access versus the right to privacy:

- If you do your work on a computer owned by someone else, they may have the right to claim access to your data files.
- If you use online services such as Facebook you should be aware that their End User License Agreement states that whatever you post belongs to them.
- If your school has an AUP for the computer lab that you have accepted, you may have allowed people other than yourself to access your data files.



Activity 5.2

1. Computer viruses often get the blame when we experience problems with our computers. Briefly explain what a computer virus is.
2. How does a computer worm work?
3. Which one of the following is NOT anti-virus software?
 - A. Avast
 - B. AVG
 - C. Sharepoint
 - D. NOD32
 - E. Chrome
 - F. Microsoft Security Essentials
 - G. Microsoft Office
4. Why is spam a type of harassment?



Something to know

Hackers can infect multiple computers in order to create a “botnet”; better known as a “zombie army”.



The computer operating system provides support software that provides security for applications. This includes preventing malicious applications, such as a keylogger, from finding out what password you enter on your banking website. Without these protections, malicious software can do incredible damage to your computer and to you as a person.

AVOIDING SECURITY THREATS

The first and best defence against security threats is to prevent them from occurring in the first place. Therefore, in order to help you avoid security threats, let's take a look at some of the following suggestions:

- Use a secure password.
- Pay careful attention to the websites and links you open.
- **Do not share your sensitive information unnecessarily:** While sharing information is part of using the internet, be selective about the websites with which you choose to share your information. If you use your email address for every website you visit, do not be surprised if you start receiving a lot of spam emails!
- **Do not open suspicious emails:** Suspicious emails are emails from unknown or suspicious email addresses, emails that seem too good to be true, emails that request sensitive information, emails with executable attachments and emails with poor spelling or grammar.
- **Do not download suspicious programs or attachments:** If you do not know who the sender of the email is, never download the attachment. Even if you know who the sender is, but you are not expecting an attachment, do not open it. Your friend may have been hacked and sending you a virus!
- **Keep your antivirus application up to date:** Your antivirus application is your last line of defence against viruses. By keeping it up to date, you make sure that it knows how to detect all the newest security threats.
- **Keep your other software up to date:** Software weaknesses or vulnerabilities can be used by hackers to gain access to your computer. One of the main reasons that programmers release updates is to fix these vulnerabilities.
- **Use two-factor authentication whenever possible:** Two-factor authentication prevents anyone from logging into your accounts using just your username and password. Instead, they need a second factor (which is usually a physical device, such as your phone) to access your account. This means that your account can only be hacked by the small number of people with access to your phone.

SAFEGUARDS AGAINST CRIMINALS, VIRUSES AND THREATS

There are times that you are not able to avoid all the security risks that we have just discussed. Fortunately, there is a solution to this problem. Firewalls and software, such as antivirus and anti-spyware, help protect users against the threats that they cannot see.

FIREWALLS

The firewall is responsible for monitoring all incoming and outgoing traffic on a network. If the firewall detects any traffic that is not within the defined set of security rules, it is blocked. Through this action, the firewall prevents malware from spreading over a computer network.

Examples of firewall software include ZoneAlarm, Comodo and GlassWire. Windows 10 comes equipped with a basic firewall, but this might not be strong enough to protect your computer against powerful malware.



Figure 5.4: A computer firewall

ANTIVIRUS PROGRAMS

An antivirus or anti-malware is a computer program designed to detect, identify, prevent and remove malware from your computer. The program achieves this by running on your computer as a background process. This allows the program to scan the computer, as well as any external storage devices that are connected to the computer, for malicious software.

Examples of antivirus programs include Bitdefender, Norton and Webroot. Windows 10 comes equipped with a basic antivirus program, called “Windows Defender”. This program should be sufficient to protect you against most threats. However, an antivirus application with more features might be needed for more serious threats.



Figure 5.5: The “Virus & threat protection” screen



Something to know

Antivirus programs were first developed to protect users against viruses, hence the name “antivirus”.



Something to know

Some anti-spyware programs are designed to only block spyware, while others are designed to block spyware, as well as other malicious programs, such as viruses.

ANTI-SPYWARE

Anti-spyware is a type of computer program designed to prevent, detect and remove spyware programs from your computer. The anti-spyware program does this by monitoring your emails, the websites you visit and files you download. If a malicious program is detected, the anti-spyware will prevent it from being transferred onto your computer. Examples of anti-spyware programs include Bitdefender, Kaspersky and AVG.



Activity 5.3

1. Your anti-virus program reports that it has quarantined a virus. What does this mean?
2. Provide THREE measures you can take to avoid computer viruses, besides installing anti-virus software and keeping the software up to date.
3. Give TWO examples of firewall software.
4. Your father says that he makes a point of updating his anti-virus software twice a year. Explain to him why this approach is not sufficient.
5. A pop-up appeared on your aunt's computer saying that she should download an anti-virus program from the internet, as viruses had been detected on the computer. Why should she not respond to this warning, if viruses are so dangerous?
6. Give the name of one popular anti-virus software package.



5.4 Impact of technology on the global community

With the invention and popularisation of the internet, the world is more connected than ever before. This connectivity has led to the creation of new technologies that have eliminated the problem of distance that we once faced. Not only have new technologies changed the way in which we find information; it has also changed the way in which we communicate, conduct business and meet new people.

COMMUNICATION

In the old days, we were limited to using landline phones to make calls and writing physical letters when communicating over a long distance. If you wanted to meet a person in another city, you had to travel to that city, or organise a meeting at a common location. Fortunately, that is no longer needed. Email, VoIP, video conferencing and smartphones have made communication much faster, cheaper and easier than ever before.

CONDUCTING BUSINESS

The internet has created a whole new platform that we can use to conduct business and to reach new customers. This includes the following:

- Creating online advertisements for products in order to reach more customers
- Creating and hosting online services, for example Netflix and Amazon
- Creating completely new business ideas, for example YouTube and Esports

MEETING NEW PEOPLE

For today's teens, friendships can start digitally: 57% of teens have met a new friend online. Social media and online gameplay are the most common digital venues for meeting friends. The most common spots for meeting friends online are social media sites like Facebook or Instagram, followed by playing networked video games. Girls who have met new friends online are more likely to meet them via social media, while boys are substantially more likely to meet new friends while playing games online.

Text messaging is a key component of day-to-day friend interactions: 55% of teens spend time every day texting with friends. Social media helps teens feel more connected to their friends' feelings and daily lives, and also offers teens a place to receive support from others during challenging times. For many teens, texting is the dominant way that they communicate on a day-to-day basis with their friends.

DISTRIBUTED COMPUTING POWER

One of most important ways in which technology has impacted our daily lives, is through the invention of **distributed computing power**. Distributed computing power technologies, such as cloud and grid computing, allow users to store information on the internet, as well as increase their computing power, by using a network where one computer can use the computing power and resources of other computers. This saves users both time and money, as they do not have to invest in expensive processing equipment.



REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

- 1.1** Which of the following is NOT a computer-related crime? (1)
- A. Downloading illegal software
 - B. Accessing hardware on a shared network
 - C. Stealing a network's password
 - D. Hotlinking
- 1.2** What does DoS stands for? (1)
- A. Denial-of-service
 - B. Data on server
 - C. Delivery of services
 - D. Digital office software
- 1.3** Which of the following can be used to launch DoS attacks? (1)
- A. Spyware
 - B. Zombies
 - C. Worms
 - D. Phishing
- 1.4** Which of the following is NOT a result of telecommunication technologies? (1)
- A. Meeting new people
 - b. Conducting business
 - C. Reducing pollution
 - D. Distributing computer power
- 1.5** What is the purpose of a firewall? (1)
- A. Preventing other computers on a network from accessing your computer
 - B. Removing viruses on the computer
 - C. Detecting spyware programs
 - D. Installing anti-malware

[5]

QUESTION 2: TRUE OR FALSE

Indicate if the following statements are TRUE or FALSE. Correct the statement if it is false.
Change the underlined word(s) to make the statement true.

- a. Malware is software that can be used to cause harm to the user's computer or computer network, without the user's knowledge. (1)
- b. Identity theft is when someone uses your credentials to access your personal information. (1)
- c. You can prevent adware by being careful when installing new applications. (1)
- d. All bots are malicious. (1)

[4]





QUESTION 3: MATCHING ITEMS

Choose a term or concept from Column B that matches a description in Column A.

(5)

COLUMN A	COLUMN B
3.1 Katya received an SMS claiming that she won a cash prize for a lucky draw she does not recall entering. The message asked her to provide her banking details so that they can transfer the money.	A. DoS B. Spoofing attacks C. Worm D. Phishing E. Trojan F. Sniffer attacks G. "Congratulations, you have won" H. Antivirus I. "Nigerian letter" J. Adware K. Firewall L. Zombie M. Anti-malware
3.2 Coleman is a hacker who intercepts data packets on a network and inspects these packets for useful information.	
3.3 Dillan was sent an email from someone claiming that they want him to help them wire some money to a charity organisation in India. In return for his efforts, they will give Dillan 5% of the transferred amount.	
3.4 When Rebetswe found a Trojan horse on her computer, she used software to remove it from her computer and check for any other viruses.	
3.5 Software that monitors the behaviour of a network and blocks any internet attacks.	

[5]

QUESTION 4: FILL IN THE MISSING WORDS

Fill in the missing word(s) in the following statements. Provide only ONE word for each space.

- a.** Businesses create _____ for products in order to reach more customers. (1)
- b.** _____ technologies saves users both _____ and money, as they do not have to invest in buying expensive processing equipment. (2)
- c.** People can use _____ to meet new people from anywhere in the world. (1)
- d.** Cell phones allow people to communicate over _____ distances and, therefore, save people money. (1)

[5]

QUESTION 5: SHORT QUESTIONS

State if the following are internet attacks or not.

(5)

DESCRIPTION	YES/NO
5.1 Phishing	
5.2 Hotlinking	
5.3 Spoofing attacks	
5.4 DoS	
5.5 Spyware	

[5]





PART 6: SCENARIO-BASED QUESTIONS

- 6.1** Carlos recently downloaded a game online. Everything seemed fine at first, until he discovered a new hidden file on his computer. He soon started to notice that his hard-drive memory was starting to get used up, even though he was not installing any large files on his computer.
- a.** To what type of threat(s) has Carlos been exposed? (2)
 - b.** What can Carlos use to get rid of this threat(s)? (1)
 - c.** Suggest three precautionary measures Carlos could take to safeguard himself against internet attacks. (3)
 - d.** Name five things Carlos could do in the future to protect himself from this type of threat(s). (5)
- 6.2** Mary downloads a lot of programs as a programmer and animator. Since Mary creates and maintains content for professional companies, she worries that her client's private information could be at risk.
- a.** What would be the best safeguard for her current situation? (1)
 - b.** Mention three benefits of the safeguard that you suggested. (3)

[15]

TOTAL: [40]

AT THE END OF THE CHAPTER

NO.	CAN YOU ...	YES	NO
1.	Explain computer-related crimes, attacks and frauds?		
2.	Describe security problems?		
3.	Identify how to avoid security threats?		
4.	List and describe safeguards against cyber attacks?		

